

CSE 385: Analysis of Algorithms (Honors)

Lecture 20 (Randomized Algorithms & High Probability Bounds)

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Markov's Inequality

Theorem 1: Let X be a random variable that assumes only nonnegative values. Then for all $\delta > 0$,

$$\Pr[X \geq \delta] \leq \frac{E[X]}{\delta}.$$

Proof: For $\delta > 0$, let

$$I = \begin{cases} 1 & \text{if } X \geq \delta; \\ 0 & \text{otherwise.} \end{cases}$$

Since $X \geq 0$, $I \leq \frac{X}{\delta}$.

We also have, $E[I] = \Pr[I = 1] = \Pr[X \geq \delta]$.

Then $\Pr[X \geq \delta] = E[I] \leq E\left[\frac{X}{\delta}\right] \leq \frac{E[X]}{\delta}$.

Example: Coin Flipping

Let us bound the probability of obtaining at least $\frac{3n}{4}$ heads in a sequence of n fair coin flips.

Let

$$X_i = \begin{cases} 1 & \text{if the } i\text{th coin flip is heads;} \\ 0 & \text{otherwise.} \end{cases}$$

Then the number of heads in n flips, $X = \sum_{i=1}^n X_i$.

We know, $E[X_i] = \Pr[X_i = 1] = \frac{1}{2}$.

Hence, $E[X] = \sum_{i=1}^n E[X_i] = \frac{n}{2}$.

Then applying Markov's inequality,

$$\Pr\left[X \geq \frac{3n}{4}\right] \leq \frac{E[X]}{3n/4} = \frac{n/2}{3n/4} = \frac{2}{3}.$$

Chebyshev's Inequality

Theorem 2: For any $\delta > 0$,

$$\Pr[|X - E[X]| \geq \delta] \leq \frac{\text{Var}[X]}{\delta^2}.$$

Proof: Observe that $\Pr[|X - E[X]| \geq \delta] = \Pr[(X - E[X])^2 \geq \delta^2]$.

Since $(X - E[X])^2$ is a nonnegative random variable, we can use Markov's inequality,

$$\Pr[(X - E[X])^2 \geq \delta^2] \leq \frac{E[(X - E[X])^2]}{\delta^2} = \frac{\text{Var}[X]}{\delta^2}.$$

Example: n Fair Coin Flips

$$X_i = \begin{cases} 1 & \text{if the } i\text{th coin flip is heads;} \\ 0 & \text{otherwise.} \end{cases}$$

Then the number of heads in n flips, $X = \sum_{i=1}^n X_i$.

We know, $E[X_i] = \Pr[X_i = 1] = \frac{1}{2}$ and $E[(X_i)^2] = E[X_i] = \frac{1}{2}$.

Then $\text{Var}[X_i] = E[(X_i)^2] - (E[X_i])^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

Hence, $E[X] = \sum_{i=1}^n E[X_i] = \frac{n}{2}$ and $\text{Var}[X] = \sum_{i=1}^n \text{Var}[X_i] = \frac{n}{4}$.

Then applying Chebyshev's inequality,

$$\Pr\left[X \geq \frac{3n}{4}\right] \leq \Pr\left[|X - E[X]| \geq \frac{n}{4}\right] \leq \frac{\text{Var}[X]}{(n/4)^2} = \frac{n/4}{(n/4)^2} = \frac{4}{n}.$$

Coin Flipping and Randomized Algorithms

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Suppose we have an algorithm that is

- correct (heads) only with probability $p \in (0,1)$, and
- incorrect (tails) with probability $1 - p$.

Question: How many times should we run the algorithm to be reasonably confident that it returns at least one correct solution?

- Las Vegas Algorithm: You keep running the algorithm until you get a correct solution. What is the bound on running time?
- Monte Carlo Algorithm: You stop after a certain number of iterations no matter whether you found a correct solution or not. What is the probability that your solution is correct (or you found a solution)?

Coin Flipping and Randomized Algorithms

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Suppose we have an algorithm that is

- correct (heads) only with probability $p \in (0,1)$, and
- incorrect (tails) with probability $1 - p$.

Suppose we run the algorithm k times.

Then probability that no run produces a correct solution is $(1 - p)^k$.

$\therefore$ probability of getting at least one correct solution is $1 - (1 - p)^k$.

Set $k = \ln_{1-p} \left(\frac{n^\alpha}{c} \right)$, where $\alpha \geq 1$ and $c > 0$ is a constant.

Then the probability that at least one run produces a correct solution is $1 - (1 - p)^k = 1 - \frac{c}{n^\alpha}$.

An event Π is said to occur with high probability if $\Pr[\Pi] \geq 1 - \frac{c}{n^\alpha}$.
w.h.p.

Example: A Coloring Problem

Let S be a set of n items.

For $1 \leq l \leq k$, let $S_l \subseteq S$ such that for every pair of $i, j \in [1, k]$ with $i \neq j$, $S_i \neq S_j$ but not necessarily $S_i \cap S_j = \emptyset$.

For each $l \in [1, k]$, let $|S_l| = r$, where $k \leq 2^{r-2}$.

Problem: Color each item of S with one of two colors, **red** and **blue**, such that each S_l contains at least one **red** and one **blue** item.

Algorithm: Take each item of S and color it either **red** or **blue** independently at random (with probability $\frac{1}{2}$).

Clearly, the algorithm does not always lead to a valid coloring (i.e., satisfy the constraints given in our problem statement).

What is the probability that it produces a valid coloring?

Example: A Coloring Problem

For $1 \leq l \leq k$, let R_l and B_l be the events that all items of S_l are colored red and blue, respectively.

Then $\Pr[R_l] = \Pr[B_l] = \left(\frac{1}{2}\right)^r = 2^{-r}$ for every $l \in [1, k]$.

$$\therefore \Pr\left[\bigcup_{l=1}^k R_l\right] = \Pr\left[\bigcup_{l=1}^k B_l\right] \leq k2^{-r} \leq 2^{r-2}2^{-r} = \frac{1}{4}.$$

Thus $\Pr\left[\bigcup_{l=1}^k (R_l \cup B_l)\right] \leq 2 \times \frac{1}{4} = \frac{1}{2}.$

$$\therefore \Pr\left[\bigcap_{l=1}^k (\bar{R}_l \cap \bar{B}_l)\right] = 1 - \Pr\left[\bigcup_{l=1}^k (R_l \cup B_l)\right] \geq 1 - \frac{1}{2} = \frac{1}{2}.$$

Hence, the algorithm is correct with probability at least $\frac{1}{2}$.

To check if the algorithm has produced a correct result we simply check the items in each S_l to verify that neither R_l nor B_l holds.

Hence, we can use this simple algorithm to design a Las Vegas algorithm for solving the coloring problem!

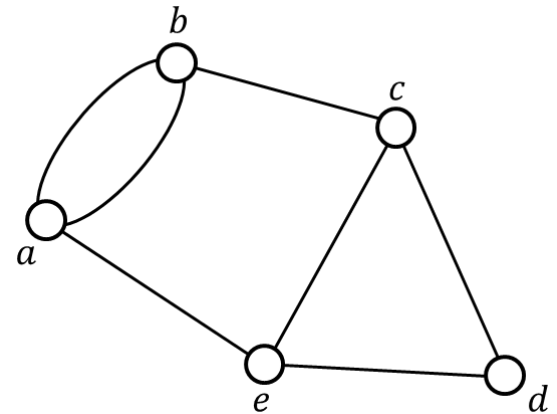
Example: The Min-Cut Problem

Let $G = (V, E)$ be a connected, undirected multigraph with $|V| = n$.

A *cut* in G is a set $C \subseteq E$, such that $G' = (V, E \setminus C)$ is not connected.

A *min-cut* is a cut of minimum cardinality.

The multigraph on the right has a min-cut of size 2: $\{(a, e), (b, c)\}$ and $\{(c, d), (d, e)\}$.



Most deterministic algorithms for finding min-cuts are based on network flows and hence are quite complicated.

Instead in this lecture we will look at a very simple probabilistic algorithm that finds min-cuts with some probability $p > 0$.

Example: The Min-Cut Problem

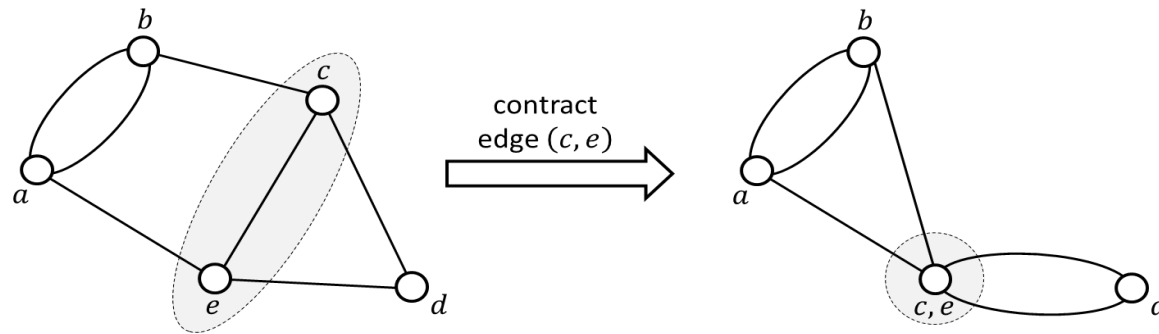
We apply the following contraction step $n - 2$ times on $G = (V, E)$:

Select an edge (say, (u, v)) from E uniformly at random.

Merge u and v into a single super vertex.

Remove all edges between u and v from E .

If as a result of the contraction there are more than one edges between some pairs of super vertices retain them all.



Let the initial graph be $G_0 = (V_0, E_0)$, where $V_0 = V$ and $E_0 = E$.

Let $G_i = (V_i, E_i)$ be the multigraph after step $i \in [1, n - 2]$.

Then clearly, $|V_i| = n - i$ and thus $|V_{n-2}| = 2$.

We return E_{n-2} as our solution.

Example: The Min-Cut Problem

Let us fix our attention on a particular min-cut C of G .

What is the probability that $E_{n-2} = C$?

Suppose $|C| = k$.

Then each vertex of $G_0 = G$ must have degree at least k as otherwise G_0 can be disconnected by removing fewer than k edges.

Hence, $|E| = |E_0| \geq k|V_0|/2 = kn/2$.

Let Π_i be the event of not picking an edge of C for contraction in step $i \in [1, n-2]$.

Then clearly, $\Pr[\Pi_1] = 1 - \frac{k}{|E_0|} \geq 1 - \frac{k}{kn/2} = 1 - \frac{2}{n}$

Also $\Pr[\Pi_2|\Pi_1] = 1 - \frac{k}{|E_1|} \geq 1 - \frac{k}{k(n-1)/2} = 1 - \frac{2}{n-1}$

In general, $\Pr[\Pi_i | \cap_{j=1}^{i-1} \Pi_j] = 1 - \frac{k}{|E_{i-1}|} \geq 1 - \frac{k}{k(n-i+1)/2} = 1 - \frac{2}{n-i+1}$

Example: The Min-Cut Problem

The probability that no edge of C was ever picked by the algorithm is:

$$\Pr\left[\bigcap_{i=1}^{n-2} \Pi_i\right] \geq \prod_{i=1}^{n-2} \left(1 - \frac{2}{n-i+1}\right) = \frac{2}{n(n-1)} > \frac{2}{n^2}.$$

So $\Pr[E_{n-2} = C] > \frac{2}{n^2}$, and $\Pr[E_{n-2} \neq C] < 1 - \frac{2}{n^2}$.

Suppose we run the algorithm $n^2/2$ times, and return the smallest cut, say C' , obtained from those $n^2/2$ attempts.

$$\text{Then } \Pr[C' \neq C] < \left(1 - \frac{2}{n^2}\right)^{n^2/2} < \frac{1}{e} \Rightarrow \Pr[C' = C] > 1 - \frac{1}{e}.$$

Hence, the algorithm will return a min-cut with probability $> 1 - \frac{1}{e}$.

But we do not know how to detect if the cut returned by the algorithm is, indeed, a min-cut.

Still we can design a Monte-Carlo algorithm based on this simple idea to produce a min-cut with high probability!

Chernoff Bounds 1, 2 and 3

Theorem 3: Let $X_1, \dots, X_n$ be independent Poisson trials, that is, each X_i is a 0-1 random variable with $\Pr[X_i = 1] = p_i$ for some p_i . Let $X = \sum_{i=1}^n X_i$ and $\mu = E[X]$. Then for any $\delta > 0$,

$$\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Corollary 1: For $0 < \gamma < \mu$, $\Pr[X \geq \mu + \gamma] \leq e^{-\frac{\gamma^2}{3\mu}}$.

Example: n Fair Coin Flips

$$X_i = \begin{cases} 1 & \text{if the } i\text{th coin flip is heads;} \\ 0 & \text{otherwise.} \end{cases}$$

Then the number of heads in n flips, $X = \sum_{i=1}^n X_i$.

We know, $E[X_i] = \Pr[X_i = 1] = \frac{1}{2}$.

Hence, $\mu = E[X] = \sum_{i=1}^n E[X_i] = \frac{n}{2}$.

Now putting $\delta = \frac{1}{2}$ in Chernoff bound 2, we have,

$$\Pr\left[X \geq \frac{3n}{4}\right] \leq e^{-\frac{n}{24}} = \frac{1}{e^{\frac{n}{24}}}.$$

Chernoff Bounds 4, 5 and 6

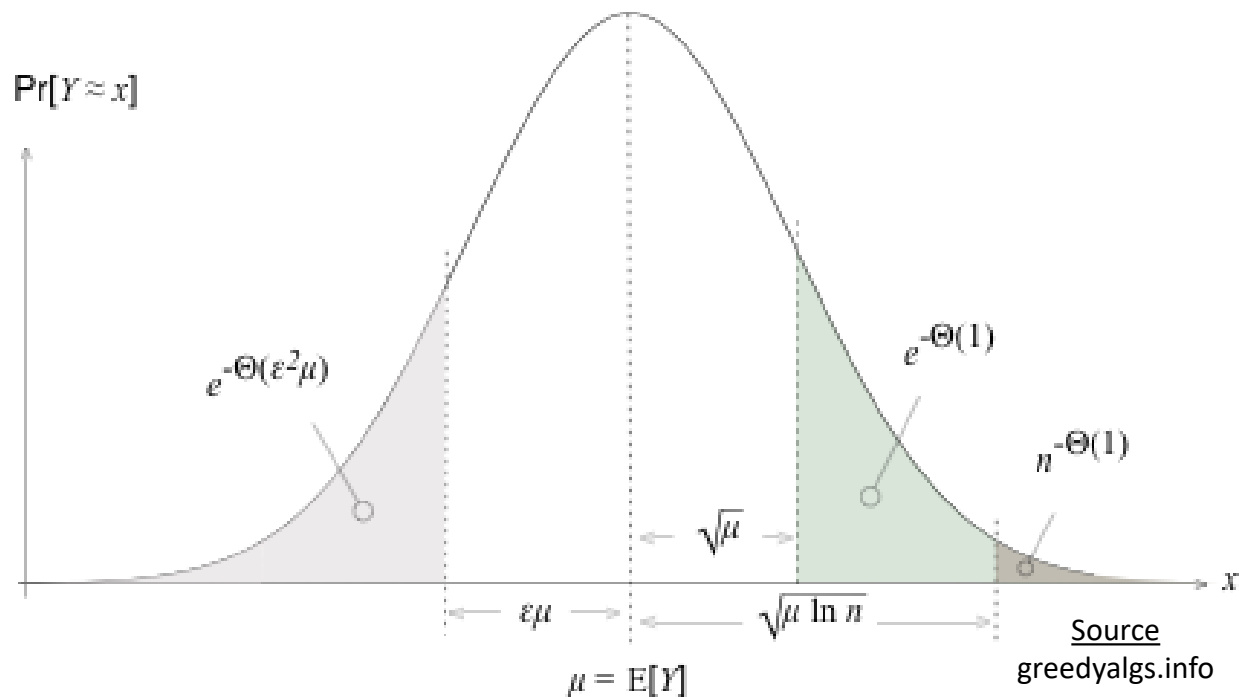
Theorem 5: For $0 < \delta < 1$, $\Pr[X \leq (1 - \delta)\mu] \leq \left(\frac{e^{-\delta}}{(1-\delta)^{(1-\delta)}} \right)^\mu$.

Theorem 6: For $0 < \delta < 1$, $\Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\mu\delta^2}{2}}$.

Corollary 2: For $0 < \gamma < \mu$, $\Pr[X \leq \mu - \gamma] \leq e^{-\frac{\gamma^2}{2\mu}}$.

Chernoff Bounds

Lower Tail	Upper Tail
$0 < \delta < 1: \Pr[X \leq (1 - \delta)\mu] \leq \left(\frac{e^{-\delta}}{(1 - \delta)^{(1-\delta)}} \right)^\mu$	$\delta > 0: \Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1 + \delta)^{(1+\delta)}} \right)^\mu$
$0 < \delta < 1: \Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\mu\delta^2}{2}}$	$0 < \delta < 1: \Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$
$0 < \gamma < \mu: \Pr[X \leq \mu - \gamma] \leq e^{-\frac{\gamma^2}{2\mu}}$	$0 < \gamma < \mu: \Pr[X \geq \mu + \gamma] \leq e^{-\frac{\gamma^2}{3\mu}}$



Randomized Quicksort **(RANDQS)**

Randomized Quicksort (RANDQS)

Input: A set of numbers S . (i.e., all numbers are distinct)

Output: The numbers of S sorted in increasing order.

Steps:

1. **Pivot Selection:** Select a number $x \in S$ uniformly at random.
2. **Partition:** Compare each number of S with x , and determine sets $S_l = \{y \in S \mid y < x\}$ and $S_r = \{y \in S \mid y > x\}$.
3. **Recursion:** Recursively sort S_l and S_r .
4. **Output:** Output the sorted version of S_l , followed by x , followed by the sorted version of S_r .

Randomized Quicksort (RANDQS)

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3. **Recursion:** Recursively sort S_l and S_r .
4. **Output:** Output the sorted version of S_l , followed by x , followed by the sorted version of S_r .

Assumption: RANDQS is called only on nonempty S .

Observation: If $|S| = n$, fewer than n recursive calls to RANDQS will be made during the sorting of S . (why?)

Observation: If $|S| = n$, and X is the total number of comparisons made in step 2 (Partition) across all (original and recursive) calls to RANDQS, then RANDQS sorts S in $O(n + X)$ time.

Expected Running Time of RANDQS

Observation: If $|S| = n$, and X is the total number of comparisons made in step 2 (Partition) across all (original and recursive) calls to RANDQS, then RANDQS sorts S in $O(n + X)$ time.

Then all we need to do is determine $E[X]$.

Let $s_1, s_2, \dots, s_n$ be the elements of S in sorted order.

Let $S_{ij} = \{s_i, s_{i+1}, \dots, s_j\}$ for all $1 \leq i < j \leq n$.

Observe that each pair of elements of S is compared at most once during the entire execution of the algorithm. (why?)

For $1 \leq i < j \leq n$, let $X_{ij} = \begin{cases} 1 & \text{if } s_i \text{ is compared to } s_j; \\ 0 & \text{otherwise.} \end{cases}$

Then $X = \sum_{i=1}^{n-1} \sum_{j=i+1}^n X_{ij}$.

Expected Running Time of RANDQS

For $1 \leq i < j \leq n$, let $X_{ij} = \begin{cases} 1 & \text{if } s_i \text{ is compared to } s_j; \\ 0 & \text{otherwise.} \end{cases}$

$$\text{Then } X = \sum_{i=1}^{n-1} \sum_{j=i+1}^n X_{ij}$$

$$\begin{aligned} \Rightarrow E[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n E[X_{ij}] \\ &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \Pr[X_{ij} = 1] \end{aligned}$$

Observations:

$X_{ij} = 0$: Once a pivot x with $s_i < x < s_j$ is chosen, s_i and s_j will never be compared at any subsequent time. (why?)

$X_{ij} = 1$: If either s_i or s_j is chosen as a pivot before any other item in S_{ij} then s_i will be compared with s_j . (why?)

Expected Running Time of RANDQS

Since each element of S_{ij} is equally likely to be chosen as a pivot:

$$\Pr[X_{ij} = 1] \leq \frac{1}{j-i+1} + \frac{1}{j-i+1} = \frac{2}{j-i+1}.$$

$$\begin{aligned} \text{Hence, } E[X] &= \sum_{i=1}^{n-1} \sum_{j=i+1}^n \Pr[X_{ij} = 1] \\ &\leq \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{2}{j-i+1} \\ &= \sum_{i=1}^{n-1} \sum_{k=1}^{n-i} \frac{2}{k+1} \\ &< \sum_{i=1}^{n-1} \sum_{k=1}^n \frac{2}{k} \\ &= \sum_{i=1}^{n-1} O(\log n) \\ &= O(n \log n) \end{aligned}$$

Thus expected running time of RANDQS is $O(n \log n)$.

High Probability Bound for RANDQS

Input: A set of numbers S . (i.e., all numbers are distinct)

Output: The numbers of S sorted in increasing order.

Steps:

1. **Pivot Selection:** Select a number $x \in S$ uniformly at random.
2. **Partition:** Compare each number of S with x , and determine sets $S_l = \{y \in S \mid y < x\}$ and $S_r = \{y \in S \mid y > x\}$.
3. **Recursion:** Recursively sort S_l and S_r .
4. **Output:** Output the sorted version of S_l , followed by x , followed by the sorted version of S_r .

We will prove that w.h.p. the running time of RANDQS does not exceed its expected running time by more than a constant factor.

In other words, we show that w.h.p. RANDQS runs in $O(n \log n)$ time.

High Probability Bound for RANDQS

Input: A set of numbers S . (i.e., all numbers are distinct)

Output: The numbers of S sorted in increasing order.

Steps:

1. **Pivot Selection:** Select a number $x \in S$ uniformly at random.
2. **Partition:** Compare each number of S with x , and determine sets $S_l = \{y \in S \mid y < x\}$ and $S_r = \{y \in S \mid y > x\}$.
3. **Recursion:** Recursively sort S_l and S_r .
4. **Output:** Output the sorted version of S_l , followed by x , followed by the sorted version of S_r .

Let us fix an element z in the original input set of size n .

We will trace the partition containing z for $c \ln n$ levels of recursion, where c is a constant to be determined later.

If a partitioning step divides S such that $\frac{|S|}{4} \leq |S_l|, |S_r| \leq \frac{3|S|}{4}$, we call that partition a *balanced* partition.

High Probability Bound for RANDQS

We will prove that among the $c \ln n$ partitioning steps z undergoes, w.h.p. at least $\frac{c}{4} \ln n$ results in balanced partitions.

If at any point z is in a partition of size k , after a balanced partitioning step it ends up in a partition of size at most $\left(\frac{3}{4}\right) k$.

Since the input size is n , after $\frac{c}{4} \ln n$ balanced partitions, z will end up in a partition of size $\leq \left(\frac{3}{4}\right)^{\frac{c}{4} \ln n} n = \frac{n}{n^{\frac{c}{4} \ln(\frac{4}{3})}}$, which is ≤ 1 for $c \geq 14$.

That means if $c \geq 14$, then z will end up in its final sorted position in the output after undergoing $\frac{c}{4} \ln n$ balanced partitions.

High Probability Bound for RANDQS

For $1 \leq i \leq c \ln n$, let

$$Z_i = \begin{cases} 1 & \text{if the partition at recursion level } i \text{ is balanced;} \\ 0 & \text{otherwise.} \end{cases}$$

But a balanced partition is obtained by choosing a pivot with rank between $\frac{k}{4}$ and $\frac{3k}{4}$, where k is the size of the set being partitioned.

Since each element of the set is chosen as a pivot uniformly at random, a balancing pivot will be chosen with probability $\frac{\frac{3k}{4} - \frac{k}{4}}{k} = \frac{1}{2}$.

$$\text{Hence, } \Pr[Z_i = 1] = \frac{1}{2}.$$

$$\text{Thus } E[Z_i] = \Pr[Z_i = 1] = \frac{1}{2}.$$

High Probability Bound for RANDQS

Total number of balanced partitions, $Z = \sum_{i=1}^{c \ln n} Z_i$.

$$\text{Then } \mu = E[Z] = \sum_{i=1}^{c \ln n} E[Z_i] = \frac{c \ln n}{2}.$$

Now applying Chernoff bound 5 (see Theorem 6) with $\delta = \frac{1}{2}$,

$$\Pr[Z \leq (1 - \delta)\mu] \leq e^{-\frac{\mu\delta^2}{2}}$$

$$\Rightarrow \Pr\left[Z \leq \frac{c}{4} \ln n\right] \leq e^{-\frac{\mu\delta^2}{2}} = e^{-\frac{c}{16} \ln n} = n^{-\frac{c}{16}} = \frac{1}{n^{\frac{c}{16}}}.$$

For $c = 32$, we have $\Pr[Z \leq 8 \ln n] \leq \frac{1}{n^2}$.

This means that the probability that z fails to reach its final sorted position even after $32 \ln n$ levels of recursion is $\leq \frac{1}{n^2}$.

High Probability Bound for RANDQS

The probability that at least one of n input elements fails to reach its final sorted position after $32 \ln n$ levels of recursion is $\leq n \times \frac{1}{n^2} = \frac{1}{n}$.

$\therefore$ the probability that all n input elements reach their final sorted positions after $32 \ln n$ levels of recursion is $\geq 1 - \frac{1}{n}$.

But observe that the total amount of work done in each level of recursion is $O(n)$.

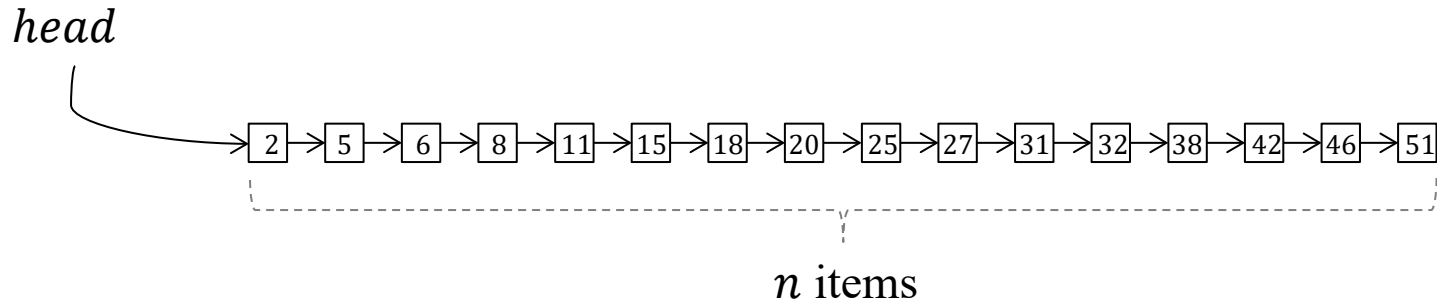
$\therefore$ total work done in $32 \ln n$ levels of recursion is $O(n \log n)$.

Hence, w.h.p. RANDQS terminates in $O(n \log n)$ time.

Random Skip Lists

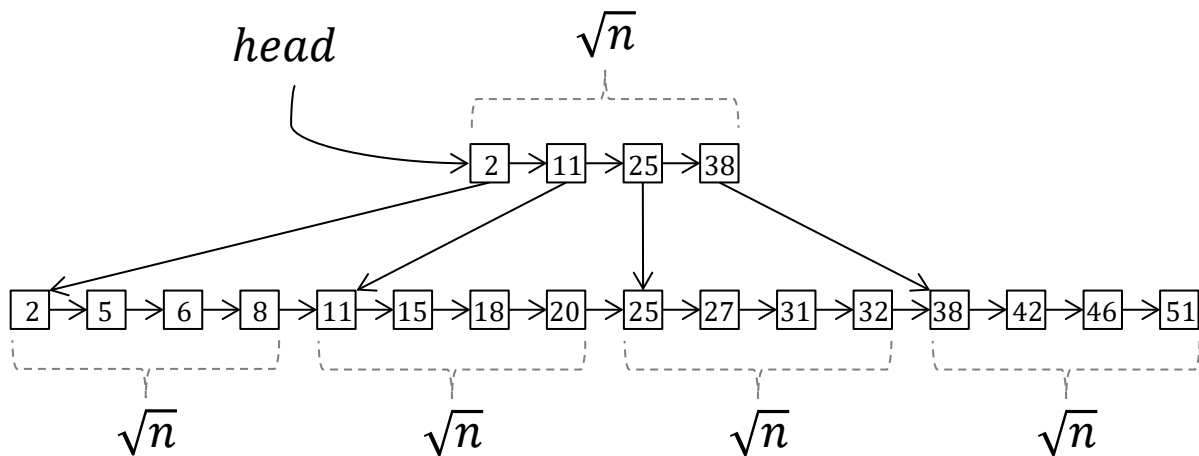
Searching in a Sorted Linked List

Traditional Linked List:



SEARCH(x): Takes $\leq n$ time.

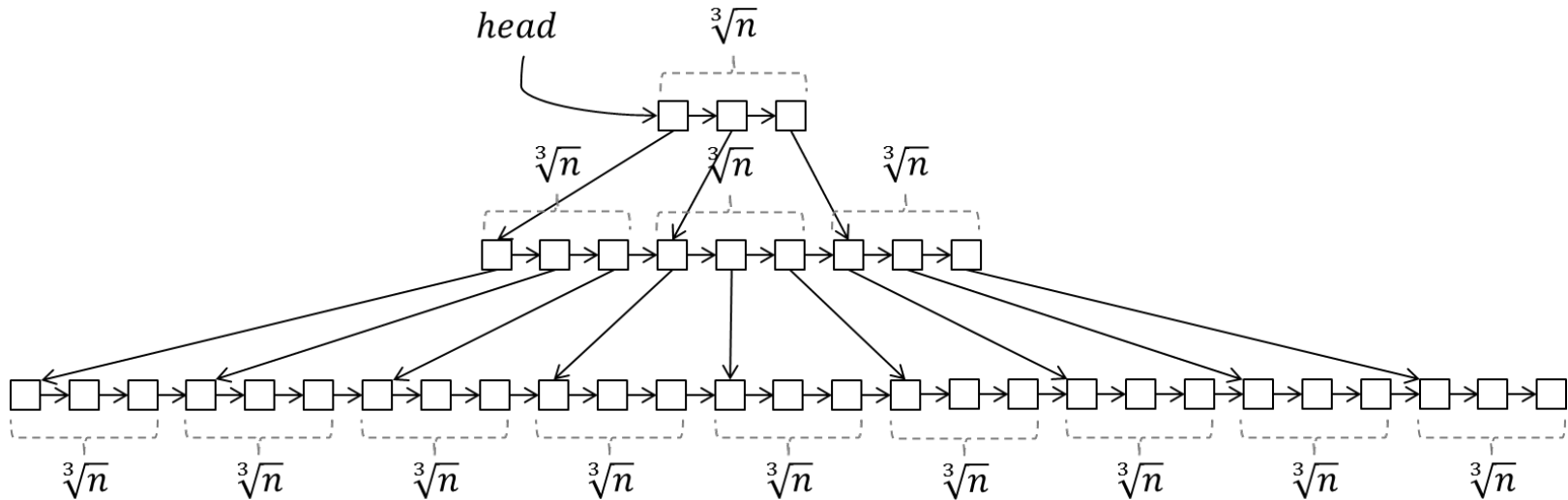
2-level Linked List:



SEARCH(x): Takes $\leq 2\sqrt{n}$ time.

Searching in a Sorted Linked List

3-level Linked List:



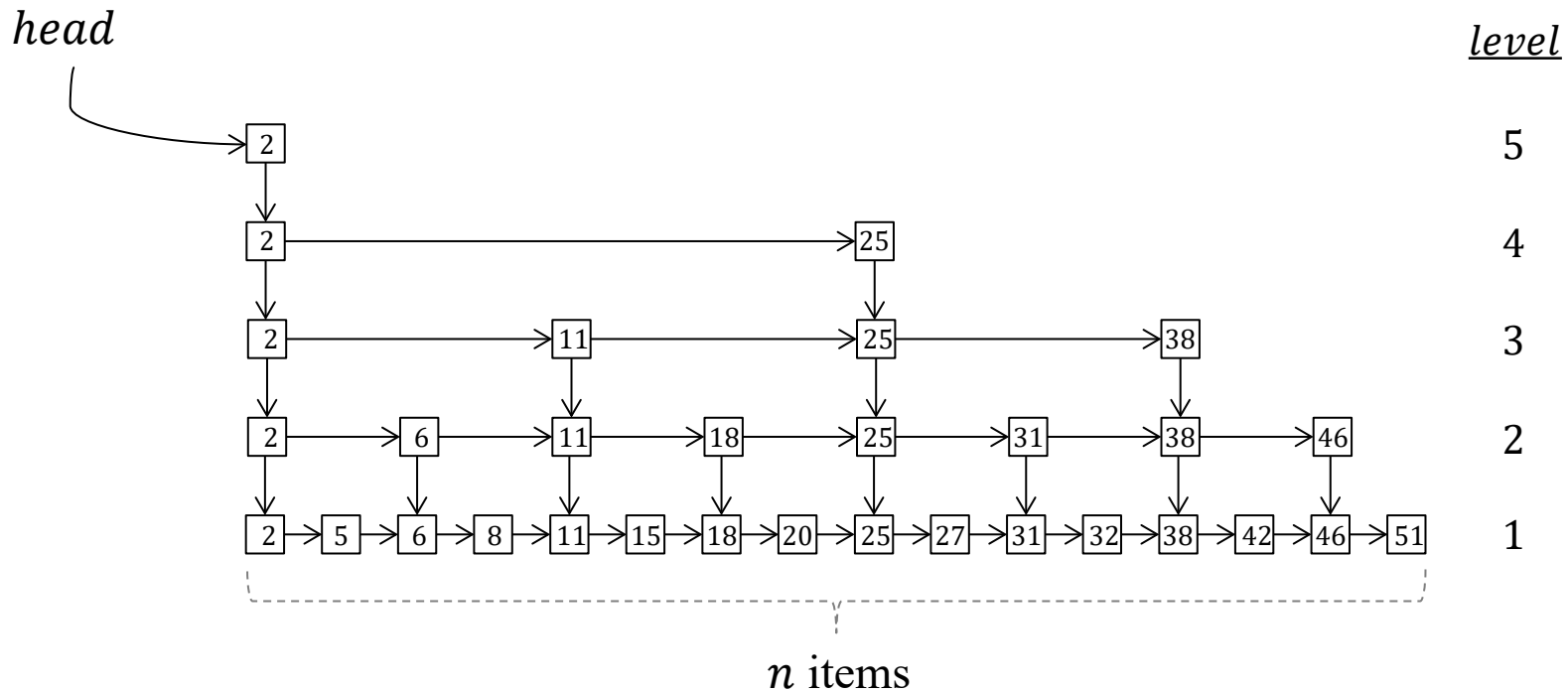
SEARCH(x): Takes $\leq 3\sqrt[3]{n}$ time.

k -level Linked List: SEARCH(x) takes $\leq k\sqrt[k]{n} = kn^{\frac{1}{k}}$ time.

For $k = \log n$: SEARCH(x) takes $\leq (\log n) \cdot n^{\frac{1}{\log n}} = 2 \log n$ time!

Searching in a Sorted Linked List

$(\log n)$ -level Linked List: SEARCH takes $\leq (\log n) \cdot n^{\frac{1}{\log n}} = 2 \log n$ time!

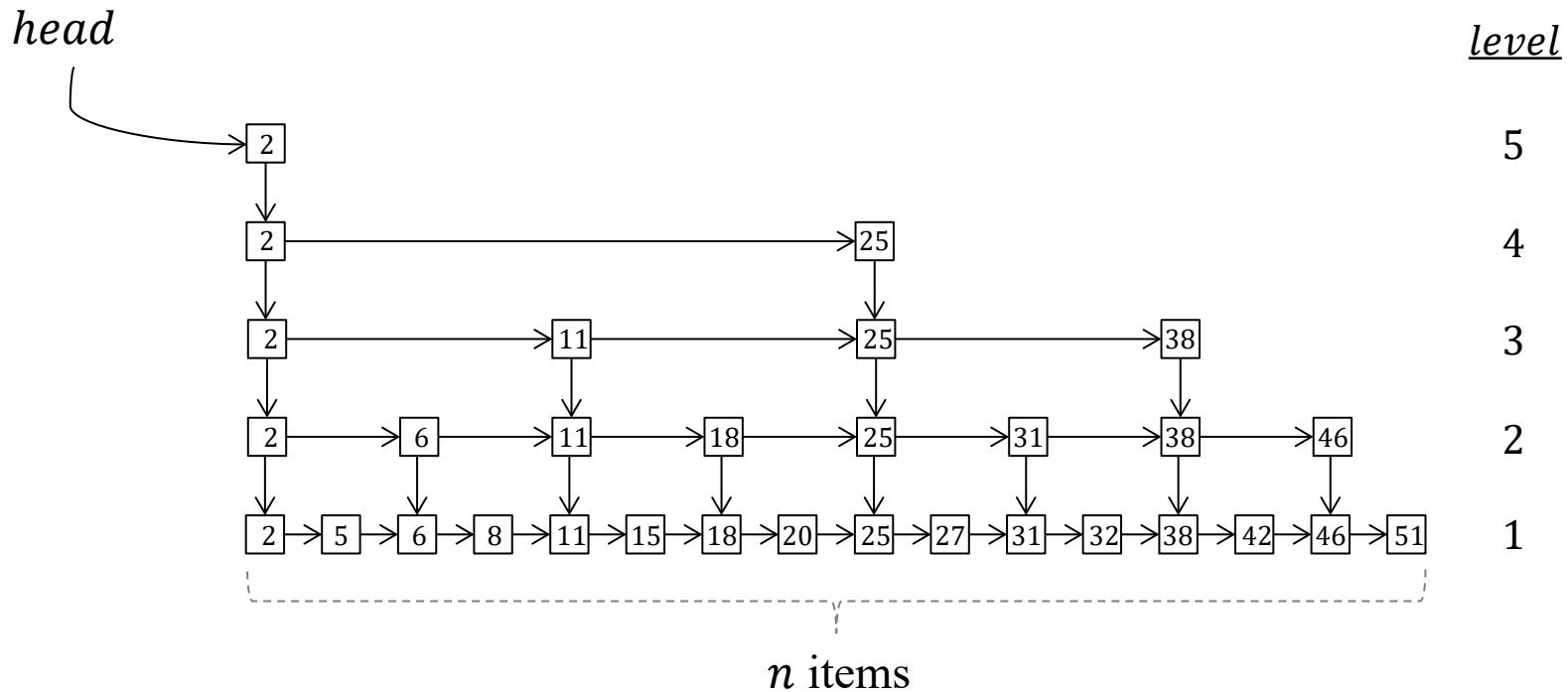


Observations:

1. Let $n_l = \text{\#items in level } l$. Then $n_{l+1} = \left\lfloor \frac{n_l}{2} \right\rfloor$.
2. Let $m_l = n_l - n_{l+1} = \text{\#items in level } l \text{ that have not reached level } l + 1$. Then $m_l = \left\lfloor \frac{n}{2^l} \right\rfloor$.

Searching in a Sorted Linked List

$(\log n)$ -level Linked List: SEARCH takes $\leq (\log n) \cdot n^{\frac{1}{\log n}} = 2 \log n$ time!

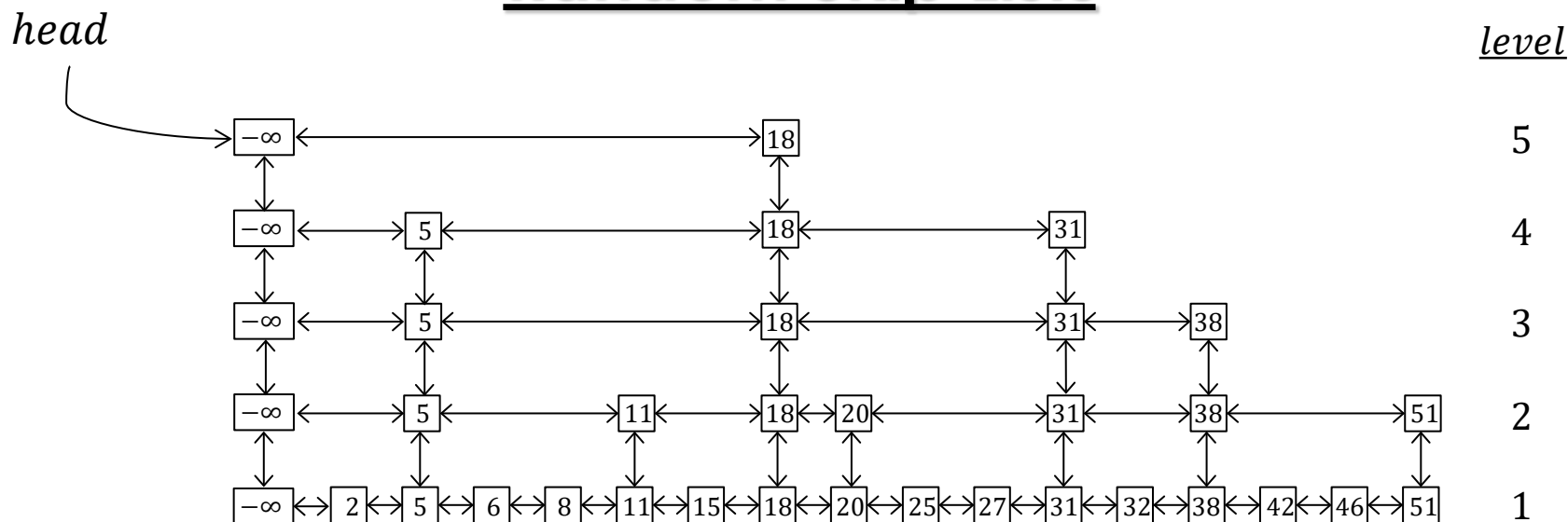


How do we maintain this regular structure under insertion and deletion of items?

Deterministic solution does not seem straightforward.

But randomization can make life really easy!

Random Skip Lists

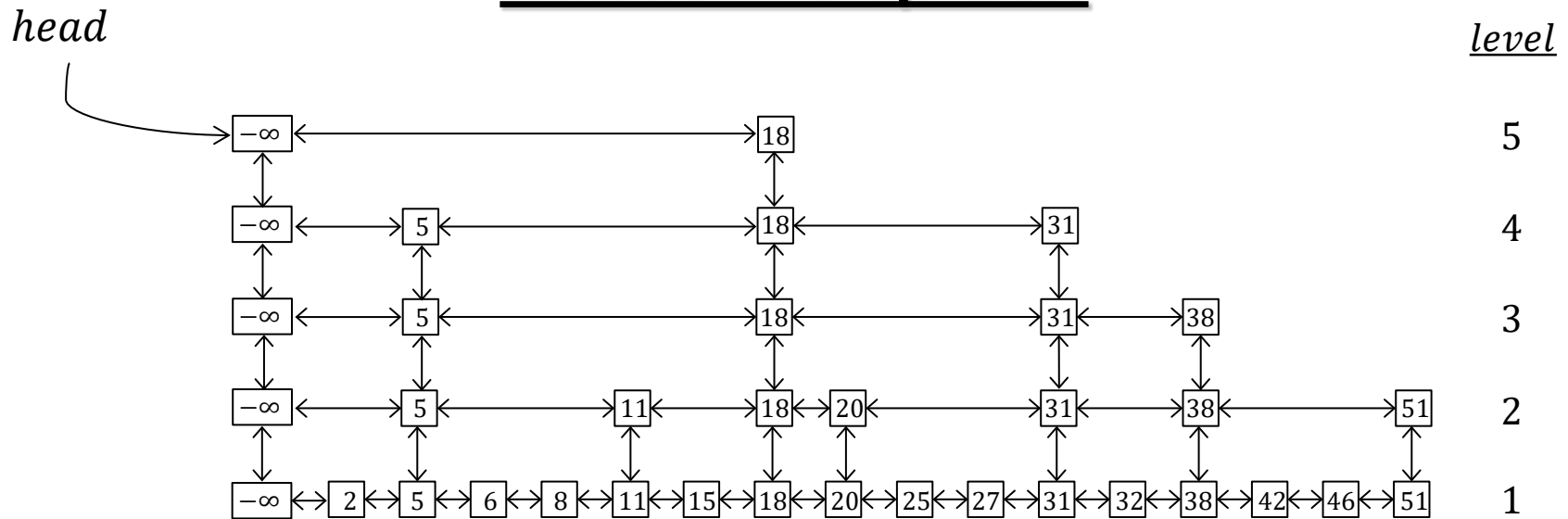


Construction:

1. Start with all items along with a sentinel $-\infty$ in level 1.
2. Promote each non-sentinel item of level $l > 0$ to level $l + 1$ with probability $\frac{1}{2}$.

If level $l + 1$ is nonempty promote the sentinel, too.

Random Skip Lists



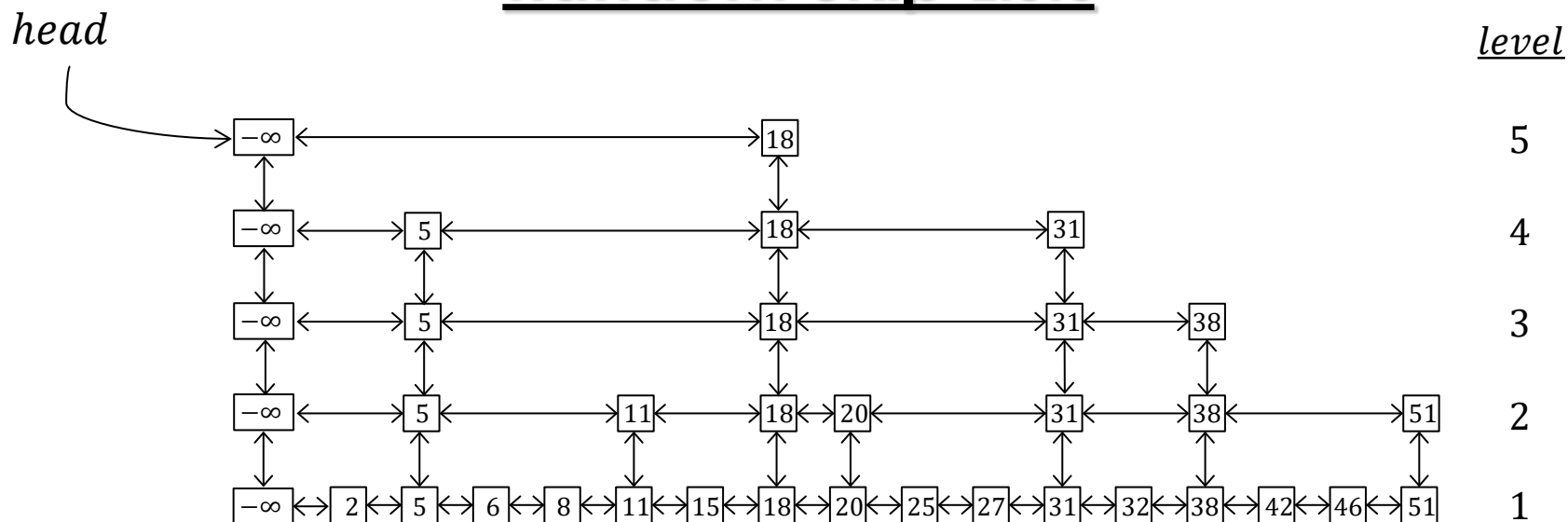
Let L be a skip list,

L_k be the set of all items in level $k \geq 1$,

$l(x) = \max\{k \mid x \in L_k\}$, and

$h(L) = \max\{l(x) \mid x \in L_0\}$.

Random Skip Lists



Clearly, for each $x \in L$ and $k \geq 1$, $\Pr[l(x) = k] = \frac{1}{2^k}$.

Then $\Pr[l(x) > k] = \sum_{i=k+1}^{\infty} \Pr[l(x) = i] = \sum_{i=k+1}^{\infty} \left(\frac{1}{2^i} \right) = \frac{1}{2^k}$.

$\therefore \Pr[h(L) \leq k] = \prod_{x \in L} \Pr[l(x) \leq k] = \left(1 - \frac{1}{2^k} \right)^n$.

$\therefore$ For constant $c > 2$, $\Pr[h(L) \leq c \log n]$

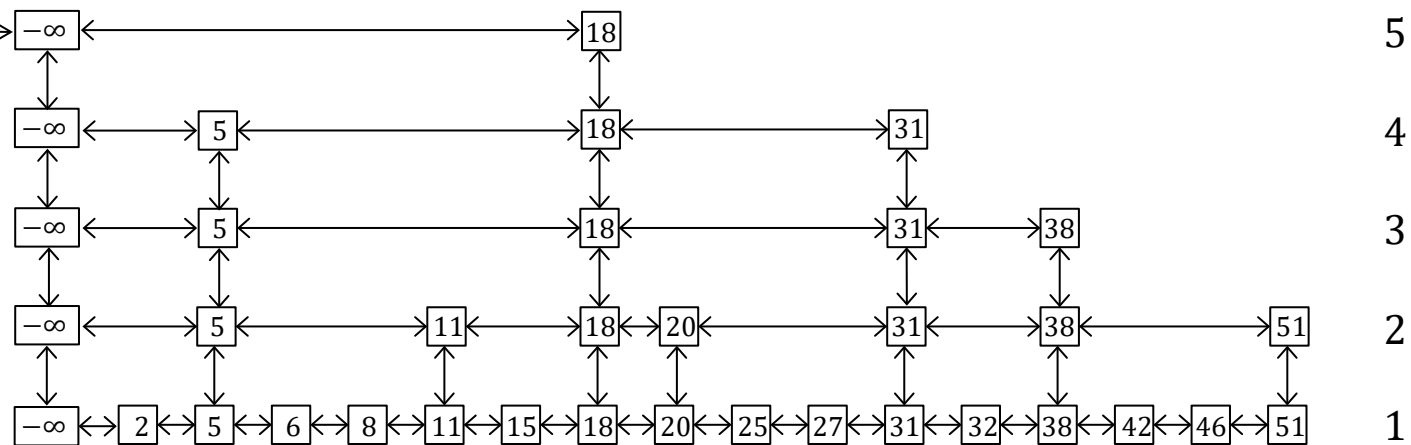
$$= \left(1 - \frac{1}{2^{c \log n}} \right)^n = \left(1 - \frac{1}{n^c} \right)^n \geq 1 - \frac{n}{n^c} = 1 - \frac{1}{n^{c-1}}.$$

Hence, w.h.p. height of a skip list is $O(\log n)$.

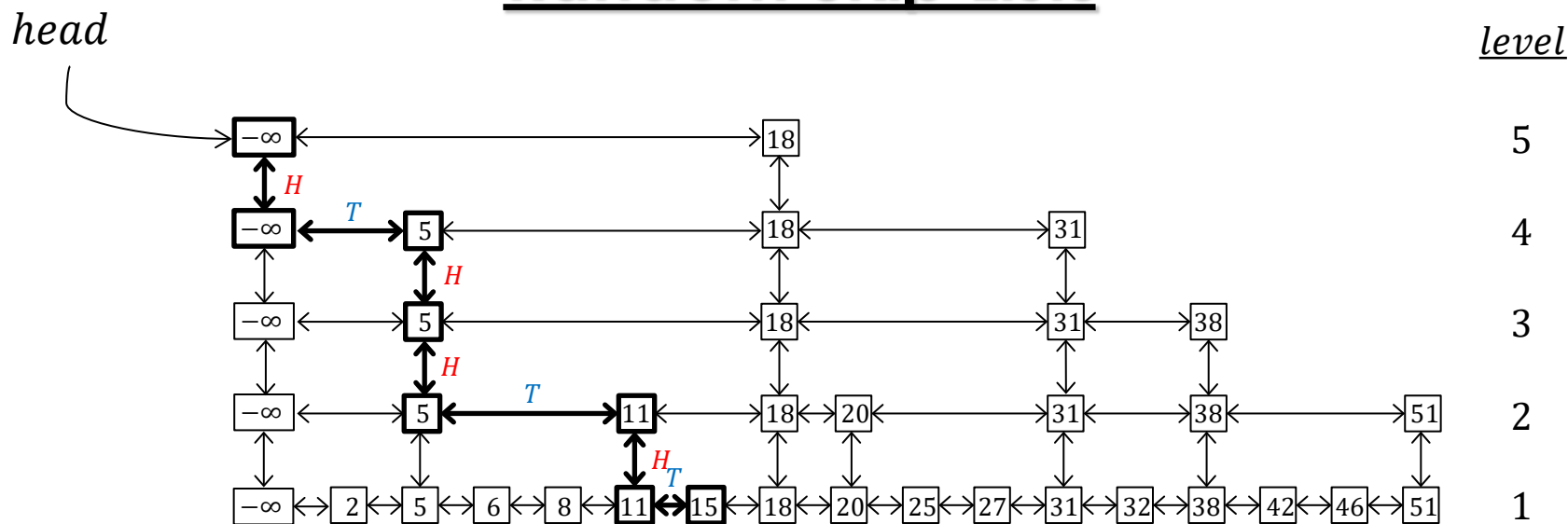
Random Skip Lists

head

level



Random Skip Lists



Let us flip $4c \log n$ fair coins and let X be the number of heads we get.

Then $\mu = E[X] = (4c \log n) \times \frac{1}{2} = 2c \log n$.

We know for $0 < \delta < 1$, Chernoff bound, $\Pr[X \leq (1 - \delta)\mu] \leq e^{-\frac{\mu\delta^2}{2}}$.

Putting $\delta = \frac{1}{2}$ and $\mu = 2c \log n$, we get, $\Pr[X \leq c \log n] \leq \frac{1}{n^4}$.

For $c \geq 4$, $\Pr[X > c \log n] \geq 1 - \frac{1}{n}$.

Hence, w.h.p. we will get more than $c \log n$ heads.

[Optional] Proofs of Chernoff Bounds

Preparing for Chernoff Bounds

Lemma 1: Let $X_1, \dots, X_n$ be independent Poisson trials, that is, each X_i is a 0-1 random variable with $\Pr[X_i = 1] = p_i$ for some p_i . Let $X = \sum_{i=1}^n X_i$ and $\mu = E[X]$. Then for any $t > 0$,

$$E[e^{tX}] \leq e^{(e^t-1)\mu}.$$

Proof: $E[e^{tX_i}] = p_i e^{t \times 1} + (1 - p_i) e^{t \times 0} = p_i e^t + (1 - p_i)$
 $= 1 + p_i(e^t - 1)$

But for any y , $1 + y \leq e^y$. Hence, $E[e^{tX_i}] \leq e^{p_i(e^t-1)}$.

$$\begin{aligned} \text{Now, } E[e^{tX}] &= E\left[e^{t \sum_{i=1}^n X_i}\right] = E\left[\prod_{i=1}^n e^{tX_i}\right] = \prod_{i=1}^n E[e^{tX_i}] \\ &\leq \prod_{i=1}^n e^{p_i(e^t-1)} = e^{(e^t-1) \sum_{i=1}^n p_i} \end{aligned}$$

But $\mu = E[X] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p_i$.

Hence, $E[e^{tX}] \leq e^{(e^t-1)\mu}$.

Chernoff Bound 1

Theorem 3: Let $X_1, \dots, X_n$ be independent Poisson trials, that is, each X_i is a 0-1 random variable with $\Pr[X_i = 1] = p_i$ for some p_i . Let $X = \sum_{i=1}^n X_i$ and $\mu = E[X]$. Then for any $\delta > 0$,

$$\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

Proof: Applying Markov's inequality for any $t > 0$,

$$\begin{aligned} \Pr[X \geq (1 + \delta)\mu] &= \Pr[e^{tX} \geq e^{t(1+\delta)\mu}] \leq \frac{E[e^{tX}]}{e^{t(1+\delta)\mu}} \\ &\leq \frac{e^{(e^t-1)\mu}}{e^{t(1+\delta)\mu}} \quad [\text{Lemma 1}] \end{aligned}$$

Setting $t = \ln(1 + \delta) > 0$, i.e., $e^t = 1 + \delta$, we get,

$$\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem 3, for $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$.

We will show that for $0 < \delta < 1$, $\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \leq e^{-\frac{\delta^2}{3}}$

$$\Rightarrow \delta - (1 + \delta) \ln(1 + \delta) \leq -\frac{\delta^2}{3}$$

That is, $f(\delta) = \delta - (1 + \delta) \ln(1 + \delta) + \frac{\delta^2}{3} \leq 0$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

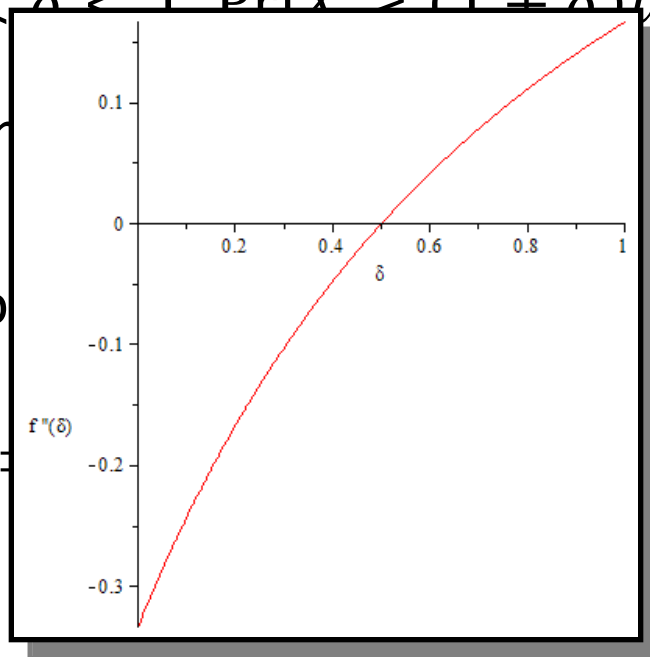
Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$ $\Pr[Y > (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem

We will show that for

That is, $f(\delta) = \delta -$



$$+ \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

$$e^{-\frac{\delta^2}{3}}$$

$$\leq -\frac{\delta^2}{3}$$

$$0$$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem 3, for $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$.

We will show that for $0 < \delta < 1$, $\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \leq e^{-\frac{\delta^2}{3}}$

$$\Rightarrow \delta - (1 + \delta) \ln(1 + \delta) \leq -\frac{\delta^2}{3}$$

That is, $f(\delta) = \delta - (1 + \delta) \ln(1 + \delta) + \frac{\delta^2}{3} \leq 0$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

Hence, $f'(\delta)$ first decreases and then increases over $[0,1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have $f'(\delta) \leq 0$ over $[0,1]$.

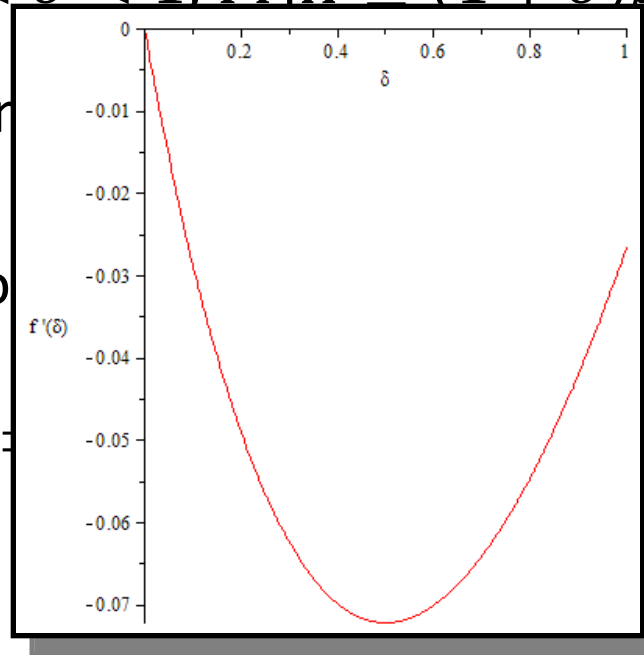
Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem

We will show that for

That is, $f(\delta) = \delta -$



$$+ \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

$$e^{-\frac{\delta^2}{3}}$$

$$\leq -\frac{\delta^2}{3}$$

$$0$$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

Hence, $f'(\delta)$ first decreases and then increases over $[0,1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have $f'(\delta) \leq 0$ over $[0,1]$.

Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem 3, for $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$.

We will show that for $0 < \delta < 1$, $\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \leq e^{-\frac{\delta^2}{3}}$

$$\Rightarrow \delta - (1 + \delta) \ln(1 + \delta) \leq -\frac{\delta^2}{3}$$

That is, $f(\delta) = \delta - (1 + \delta) \ln(1 + \delta) + \frac{\delta^2}{3} \leq 0$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

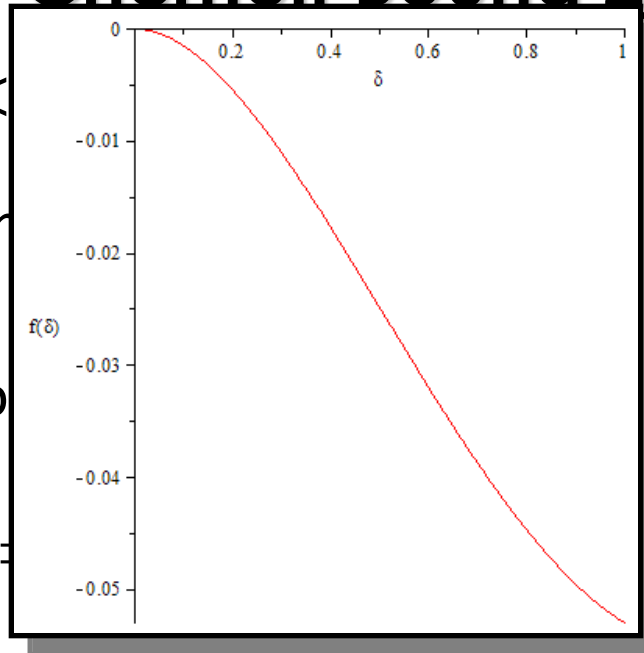
Hence, $f'(\delta)$ first decreases and then increases over $[0,1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have $f'(\delta) \leq 0$ over $[0,1]$.

Since $f(0) = 0$, it follows that $f(\delta) \leq 0$ in that interval.

Chernoff Bound 2

Theorem 4: For $0 < \delta \leq 1$,



$$] \leq e^{-\frac{\mu\delta^2}{3}}.$$

Proof: From Theorem

$$+ \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \right)^\mu.$$

We will show that for

$$e^{-\frac{\delta^2}{3}}$$

$$\leq -\frac{\delta^2}{3}$$

That is, $f(\delta) = \delta - (1 + \delta) \ln(1 + \delta) + \frac{\delta^2}{3} \leq 0$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

Hence, $f'(\delta)$ first decreases and then increases over $[0,1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have $f'(\delta) \leq 0$ over $[0,1]$.

Since $f(0) = 0$, it follows that $f(\delta) \leq 0$ in that interval.

Chernoff Bound 2

Theorem 4: For $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] \leq e^{-\frac{\mu\delta^2}{3}}$.

Proof: From Theorem 3, for $\delta > 0$, $\Pr[X \geq (1 + \delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{(1+\delta)}}\right)^\mu$.

We will show that for $0 < \delta < 1$, $\frac{e^\delta}{(1+\delta)^{(1+\delta)}} \leq e^{-\frac{\delta^2}{3}}$

$$\Rightarrow \delta - (1 + \delta) \ln(1 + \delta) \leq -\frac{\delta^2}{3}$$

That is, $f(\delta) = \delta - (1 + \delta) \ln(1 + \delta) + \frac{\delta^2}{3} \leq 0$

We have, $f'(\delta) = -\ln(1 + \delta) + \frac{2}{3}\delta$, and $f''(\delta) = -\frac{1}{1+\delta} + \frac{2}{3}$

Observe that $f''(\delta) \leq 0$ for $0 \leq \delta \leq \frac{1}{2}$, and $f''(\delta) > 0$ for $\delta > \frac{1}{2}$.

Hence, $f'(\delta)$ first decreases and then increases over $[0,1]$.

Since $f'(0) = 0$ and $f'(1) < 0$, we have $f'(\delta) \leq 0$ over $[0,1]$.

Since $f(0) = 0$, it follows that $f(\delta) \leq 0$ in that interval.

Chernoff Bound 3

Corollary 1: For $0 < \gamma < \mu$, $\Pr[X \geq \mu + \gamma] \leq e^{-\frac{\gamma^2}{3\mu}}$.

Proof: From Theorem 2, for $0 < \delta < 1$, $\Pr[X \geq (1 + \delta)\mu] < e^{-\frac{\mu\delta^2}{3}}$.

Setting $\gamma = \mu\delta$, we get, $\Pr[X \geq \mu + \gamma] \leq e^{-\frac{\gamma^2}{3\mu}}$ for $0 < \gamma < \mu$.